

PROPOSED SOLUTION

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

Rising Waters is a web-based Machine Learning application developed to predict flood risk using rainfall and weather parameters. The system helps government authorities, disaster management teams, and weather departments make timely decisions by providing accurate flood predictions.

Proposed Solution

The proposed solution is to develop a smart flood prediction system that uses Machine Learning algorithms to analyze environmental data and predict flood occurrence. The application accepts rainfall and weather-related inputs, processes them using a trained XGBoost model, and displays the flood risk prediction through an interactive web dashboard.

Working Process

Step 1:

The user opens the Rising Waters application.

Step 2:

The user enters rainfall and weather parameters such as annual rainfall, seasonal rainfall, temperature, humidity, and cloud cover.

Step 3:

The application validates the entered values and preprocesses the data.

Step 4:

The Machine Learning model analyzes the input data and predicts the flood risk.

Step 5:

The prediction result is displayed as High Flood Risk or Low Flood Risk.

Step 6:

Authorities use the prediction for disaster preparedness and evacuation planning.

Key Features

- AI-powered flood prediction.
- Accurate Machine Learning model.
- Interactive Streamlit dashboard.
- Easy-to-use interface.
- Fast prediction results.
- Early warning support.

- Reliable environmental data analysis.

Advantages

- Reduces flood-related losses.
- Supports disaster management.
- Improves emergency response.
- Provides accurate predictions.
- Saves time and human effort.
- Can be deployed online using Streamlit.

Expected Outcome

The Rising Waters application provides fast and reliable flood predictions that help authorities take preventive actions before flood situations become severe. The system improves public safety and disaster management efficiency.

Conclusion

The proposed solution combines Artificial Intelligence, Machine Learning, and environmental data to build a smart flood prediction platform. Rising Waters provides a practical and scalable solution for early flood warning and disaster preparedness.